



Designing gamified apps for sustainable consumption: A field study

Rory Mulcahy^a, Rebekah Russell-Bennett^b, Dawn Iacobucci^{c,*}

^a University of the Sunshine Coast, Hervey Bay, Australia

^b QUT Business School, Queensland University of Technology, Brisbane, Australia

^c Vanderbilt University, Nashville, United States



ARTICLE INFO

Keywords:

Gamification
Sustainability
Serious games
Gamified app
Social marketing
Sustainable consumption
Energy
Usage data

ABSTRACT

This research tests the effectiveness of a social marketing field study on sustainable consumption conducted by energy companies on real customers. The study uses a gamified app to encourage sustainable household energy usage. We analyze app usage, survey, and energy bills data to demonstrate the impact of specific game design elements on customers' perceived enjoyment and knowledge, and their behavioral intentions regarding sustainable consumption. We show the gamified app influences energy saving behaviors and word-of-mouth, and results in significant monetary savings compared to a control group. These are exciting results that provide novel insights and help support the potential of gamified apps for businesses and consumers.

1. Introduction

Climate changing effects of fossil fuel usage resulting in deforestation, water droughts, coral reef depletion, melting polar caps, and threats to wildlife are abundantly documented, both as scientific evidence (Palumbi, Barshis, Traylor-Knowles, & Bay, 2014; Urban, 2015) and in popular press coverage (Gore, 2007; WWF, 2018). Countries such as Australia and cities from Amsterdam to Miami to Venice are said to be “on the brink” of catastrophe (Ryan, 2017). Scientists urge more thoughtful, sustainable consumption by trying to change people's attitudes, values, and behaviors (Young, Hwang, McDonald, & Oates, 2009).

Sustainability is thought to be a multi-faceted concept aimed at consumer, societal, and environmental well-being (Peattie & Peattie, 2009; UNESCAP, 2015). The focus of the current research is on energy consumption which is an important business and consumer issue. If consumers reduced their household energy usage, they would enhance their financial well-being through economic savings as well as reduce pressures on energy supply and production via natural resources, thereby simultaneously enhancing both economic and environmental sustainability.

These goals comprise compatible incentives—both the household and the environment would directly and immediately benefit from less energy usage. The fact that the goals are aligned gives companies some flexibility—their messaging can be about the abstract goal of “helping the environment” or the immediate goal of “saving money.” Research has shown that while consumers can understand the importance of

sustainability goals and even feel good about participating in such efforts, appeals to self-oriented behaviors such as monetary savings and economic incentives are frequently more effective than abstract appeals such as pollution reduction (Caird, Roy, & Herring, 2008).

A separate contemporary reality is the massive size of the video game and gamified apps industries. *Bloomberg Technology* estimates the global market value of the gaming industry (for 2016) at \$101.1 billion. There is also increasing scholarly interest in games and gamification, particularly for contexts like our research, in which gamified apps are designed not just for pleasure but also for more serious purposes—in our case, increasing knowledge about energy consumption, and changing consumer energy behaviors.

This research is at the nexus of these phenomena of sustainability and gamified apps. We investigate whether the use of gamified apps for sustainable consumption of energy may accrue benefits to consumers (energy bill savings), businesses (WOM behaviors), and the environment (reduced energy consumption). Our intended contributions are threefold: 1) to understand the impact of specific game design elements on consumers' perceived value and energy saving behaviors, 2) to obtain specific diagnostic information about the precise nature of the impact of the game design elements on different energy saving behaviors, and 3) to understand app user differences. To achieve these contributions, we conducted a field study involving real customers of energy companies that utilized a gamified app to educate consumers of the connections of their actions (e.g., switching off lights) to their household energy consumption.

* Corresponding author.

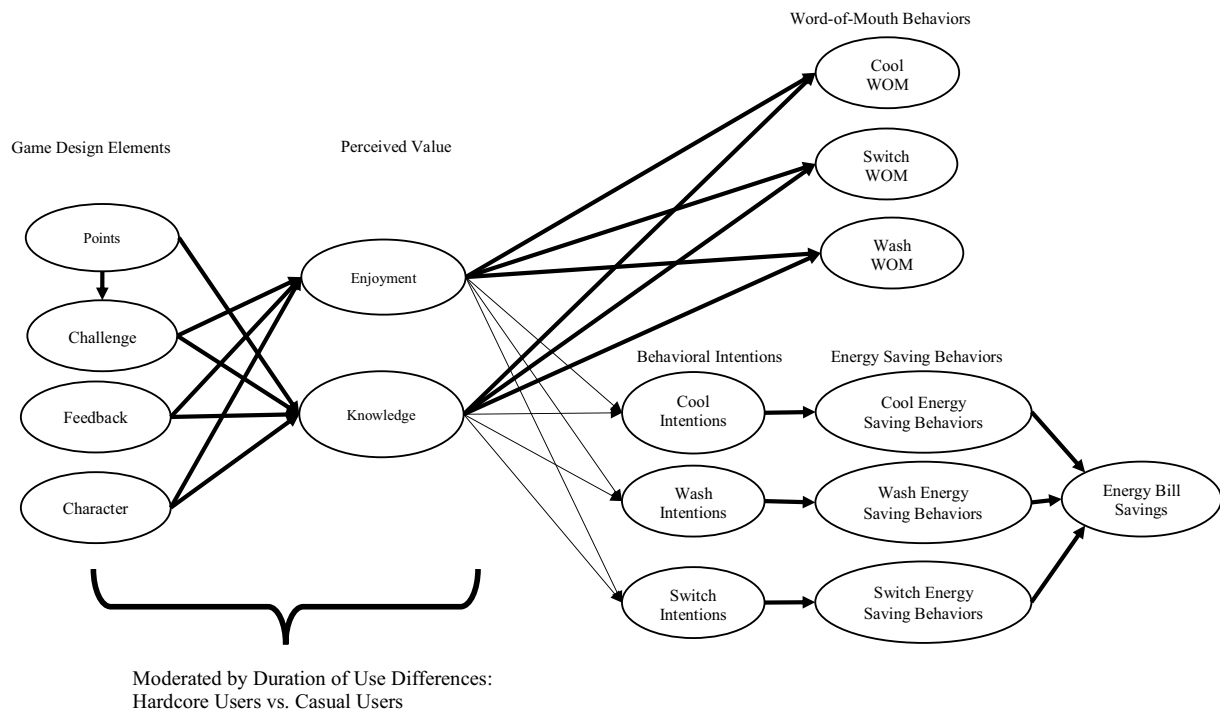
E-mail addresses: rmulcahy@usc.edu.au (R. Mulcahy), rebekah.bennett@qut.edu.au (R. Russell-Bennett), dawn.iacobucci@owen.vanderbilt.edu (D. Iacobucci).

<https://doi.org/10.1016/j.jbusres.2018.10.026>

Received 7 November 2017; Received in revised form 9 October 2018; Accepted 10 October 2018

Available online 17 October 2018

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Bold arrows depict contributions new to this research. Narrow arrows depict hypotheses replicating relationships in the literature.

Fig. 1. Conceptual model.

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The remainder of the paper is organized as follows. We present the conceptual model and the relevant literature, including the gamification literature on game design elements and the emerging area of gamification for sustainability. Next, we review the literature on consumers' subjective experiences of games and gamification, attitudes, behavioral intentions, and actual energy saving behaviors, and potential moderation by user differences. Hypotheses are then presented, followed by the study methodology and results. Finally, implications for theory and practice are discussed.

2. Conceptual model

Fig. 1 depicts the conceptual model guiding this research. Our main research contributions appear at the left regarding the specific game design elements, and the right regarding actual energy-related behaviors (not just behavioral intentions), word-of-mouth (WOM), and energy bill savings. The constructs in the center and their relationships are more established in the literature, and we test them for thoroughness.

2.1. Gamification and serious games

To begin with terminology, “gamification” has been defined as “the use of video game elements...to improve user experience and user engagement” (Deterding, Dixon, Khaled, & Nacke, 2011, p.10), whereas “serious games” has been defined as “any form of interactive computer-based game software...developed with the intention to be more than entertainment” (Ritterfeld, Cody, & Vorderer, 2009, p.6). Gamification and serious games share a common goal of capitalizing on the popularity of games with the view of changing real world behaviors (Johnson, Horton, Mulcahy, & Foth, 2017; Mulcahy, Russell-Bennett, Zainuddin, & Kuhn, 2018). The smartphone app used in our research draws on aspects from both definitions, thus in this paper, so as not to create confusion between the usage of “gamification” or “serious games,” we use the term “gamified app” and individuals who use the gamified app as “users.”

2.1.1. Gamification and game design elements

In the literature, there is a plethora of “game design elements,” including points earned, leader boards, achievements, levels, story, clear goals, feedback, rewards, progress and challenge (Johnson et al., 2017). For example, the gamified *FitBit* app provides users with challenges (steps walked per day), feedback (weight loss progress) and badges (goals achieved). Yet Johnson et al. (2017) noted that only two of 25 studies examined the effects of specific game design elements, an oversight in the literature that is surprising given that gamified apps are based on the use of game design elements to influence attitudes and behavior (Huotari & Hamari, 2017).

In our research, we examine the impact of points, challenge, feedback, and character on users' gained knowledge and enjoyment. Thus, the first intended contribution of this research is to better understand the impact of *specific game design elements* on users' knowledge, enjoyment, intentions, and energy consumption behaviors.

2.1.2. Gamification and sustainability

Some research is beginning to suggest that gamified apps are enjoyable for users and can potentially be effective in achieving some positive attitude modification toward embracing sustainable behaviors (Johnson et al., 2017). The current research will extend beyond intentions to actual energy-related behaviors.

Energy efficiency behaviors and improvements are so important that companies are exploring creative means of conveying to their customers information about energy consumption, such as providing apps that measure customers on smart grids and relay to customers' information about their electricity usage (Moreno-Munoz, Bellido-Outeirino, Siano, & Gomez-Nieto, 2016). Companies are finding positive receptivity among customers who enjoy feeling more engaged with the company, and appreciate understanding their active and decisive role in the energy bills they receive (Moreno-Munoz et al., 2016).

Madani, Pierce, and Mirchi (2017) describe one of the strengths of gamification in arenas like environmental sustainability is that they allow users to simulate scenarios that would be difficult or costly to

create in real life. Thus, users can see how their actions might impact the environment, without any detrimental results actually unfolding. In our gamified app, points are achieved when the avatar engages in energy saving behaviors, and when the user performs the behaviors in real life, doing so translates into energy bill savings.

Morganti et al. (2017) reviewed 10 studies and found the impact of gamification to be very promising, but they indicated that more research was necessary to understand the effects on real behavioral outcomes. Thus, our research examines the sequence of relationships from gamified apps toward desired behavioral outcomes (e.g., energy bill reduction), including via mediators as discussed next.

2.2. Mediators: perceived value of enjoyment and knowledge

Perceived value refers to users' experiences and motivations for engaging with a gamified app (cf., Koivisto & Hamari, 2014). Marketers have long considered hedonic (or entertaining) and utilitarian (functional) benefits, and their impact on outcomes such as customer satisfaction (Barbopoulos & Johansson, 2017). Given the multi-purposed function of gamified apps—being enjoyable to use while also having a serious functional benefit, both aspects are integrated into our research (Hamari & Koivisto, 2015; Sharma & Jha, 2017). The terminology of hedonic versus utilitarian can be imprecise, so we refer to hedonic value as “enjoyment” with the gamified app, and utilitarian value as reflecting the “knowledge” consumers gain from the app.¹

Enjoyment refers to a user's emotional pleasure interacting with a gamified app (Hsu & Lu, 2004), measured via perceived enjoyment or playfulness and escapism (Table 1). Enjoyment has been shown to influence intentions to exercise (Hamari & Koivisto, 2015), use e-banking services (Rodrigues et al., 2016) and reduce drinking (Mulcahy et al., 2015).

Knowledge in the gamification literature is represented by perceived usefulness and ease of use (Table 1). Pihlström and Brush (2008) found knowledge created from using a mobile service predicted desired outcomes. Many marketing efforts hold to the paradigm that increasing consumers' awareness or knowledge about the effects of less desirable behaviors (e.g., smoking, energy consumption) leading to maladaptive outcomes (lung cancer, global warming) will help modify consumers' perceptions, intentions and behaviors; hence we also include knowledge.

2.3. Sustainability outcomes

While showing the different effects of game design elements on mediators of enjoyment and knowledge is important (Johnson et al., 2017), this study was also designed to test the downstream effects on real behaviors—three specific energy saving behaviors (described shortly) and energy bill savings. Outcomes may be characterized as those for consumers (energy bill savings), businesses (WOM), and the environment (reduced energy consumption). Most proponents of gamification believe that apps should lead to observable behavior changes, yet some have pointed to a lack of such empirical evidence (Morganti et al., 2017).

We will examine the effectiveness of gamification on influencing multiple “lower-order” energy saving behaviors, which contribute to the performance of a “higher-order” behavior. For example, instead of an app urging consumers toward a goal such as “help save the environment!,” it may be more achievable to suggest specific measures consumers might undertake, such as “switch off lights when not in use!” Our gamified app encourages three specific behaviors labeled “Cool,” “Wash,” and “Switch” (described shortly), each of which would reduce energy usage and yield energy bill savings. The second intended

Table 1
Overview of literature.

Author (year)	Area of focus	Measurement	Behavior	Game design elements	Enjoyment (hedonic)	Knowledge (utilitarian)	Outcomes
Rodrigues, Oliveira, and Costa (2016)	Gamification	Self-report	E-banking (+)	N/A	Enjoyment	Ease of use, Usefulness	Behavioral intentions
Hamari and Koivisto (2015)	Gamification	Self-report	Exercise (+)	N/A	Playfulness, Enjoyment	Ease of use, Usefulness	Attitude, Behavioral intentions
Mulcahy, Russell-Bennett, and Rundle-Thiele (2015)	Gamification	Self-report	Alcohol consumption (-)	N/A	Playfulness	N/A	Behavioral intentions
Bittner and Shipper (2014)	Gamification	Self-report	Purchase behavior (+)	N/A	Enjoyment	Usefulness	Purchase intentions
Koivisto and Hamari (2014)	Gamification	Self-report	Exercise (+)	N/A	Playfulness, Enjoyment	Ease of Use, Usefulness	Behavioral intentions
Liu & Li (2011)	Gaming	Self-report	Gaming use (+)	N/A	Enjoyment	Usefulness, Ease of use	Attitude
Park & Lee (2011)	Gaming	Self-report	Online game purchases (+)	Character	Enjoyment	Monetary value	Behavioral intentions
Koo (2009)	Gaming	Self-report	Online gaming (+)	N/A	Visual authority	Epistemic curiosity	Purchase intentions
Hsu and Lu (2004)	Gaming	Self-report	Online gaming (+)	N/A	Enjoyment	Concentration	Behavioral intentions
This study	Gamification	Self-report, Data analytics	Household Energy usage (-)	Points, Challenge, Feedback, Character	Enjoyment	Ease of use, usefulness	Behavioral intentions, Game Usage
					Enjoyment	Knowledge	Behavioral intentions, Word-of-mouth, Energy saving behaviors, Bill savings

+ (–) = increase (decrease) behavior; Gamification focused on games for purposes beyond entertainment, gaming focused on entertainment.

¹ We thank an anonymous Reviewer for suggesting this terminology for our constructs.

contribution of this research is therefore to investigate the impact of the gamified app on lower-order specific behaviors which help comprise sustainable energy consumption.

2.4. Duration of use-hardcore versus casual users

There is also a need to understand how gamified apps affect different kinds of users (Johnson et al., 2017; Koivisto & Hamari, 2014). There are likely to be heterogeneous responses to gamified apps, such as those between “hardcore” and “casual” users, based upon their level of involvement with the app (Scharkow, Festl, Vogelgesang, & Quandt, 2015). Users who are new versus those who have logged many hours are known to seek different game design elements (Scharkow et al., 2015). For example, users who are hardcore are known to be competitive and highly involved, whereas casual users play only occasionally and are less involved (Scharkow et al., 2015). Thus, the third intended contribution of this research is to understand possible moderating effects of user differences based upon their duration of use.

3. Hypothesis development

3.1. Effects of game design elements

In this section, we present our hypotheses, beginning with the effects of specific game design elements. Our gamified app contained four game design elements: points, feedback, character, and challenge, each described next.

3.1.1. Points

The data for “points” achieved while using the gamified app were derived from app analytics. Prior research shows that accumulating points is important for general entertainment games (Yee, 2006), thus, we hope to confirm that acquiring points will drive users' feelings of enjoyment and knowledge. Points can also inform a user as to the cause-effect relationships of real-world behaviors within a virtual environment, assisting users in understanding consequences of their actions (Orji, Vassileva, & Mandryk, 2014), thus points should also affect users' perceived challenge and achievement.

3.1.2. Feedback

Feedback has been shown to be an important design consideration for e-learning games (Fu et al., 2009). We expect that feedback from the app will reinforce feelings of enjoyment and knowledge, as predicted by behavioral learning theories such as operant conditioning, whereby reinforcement influences performance. This prediction also draws on self-determination theory in that consumers have needs which include feelings of competence and autonomy, which can be affected by feedback in a gamified app (Ryan, Rigby, & Przybylski, 2006).

3.1.3. Character

One of the key needs of individuals outlined in self-determination theory is that of “relatedness.” In gamification, characters or avatars can represent the role a user takes while immersed, and a point of reference for users inside and outside of the gamified environment (Ryan et al., 2006). Thus, we will test the impact of character on users' perceived value.

3.1.4. Challenge

The “challenge” of the app represents the user's subjective experience, rather than an objective design feature, for example, a novice may earn few points yet feel that the gamified app was pleasantly challenging, whereas an expert user may achieve more points and perhaps wish the app presented even more challenge. Hamari et al. (2016) found challenge could enhance learning, and Russell-Bennett, Leo, Rundle-Thiele, and Drennan (2016) found challenge could enhance feelings of enjoyment.

Together, the game design element hypotheses follow:

H1. *Points should affect: (a) enjoyment and (b) knowledge, (c) challenge.*

H2. *“Feedback” should affect: (a) enjoyment and (b) knowledge.*

H3. *“Character” should affect: (a) enjoyment and (b) knowledge.*

H4. *“Challenge” should affect: (a) enjoyment and (b) knowledge.*

Next, we present hypotheses about the mediating process from game design elements to energy saving behaviors.

3.2. Effects of the mediator perceived value on intentions and actual behaviors

Our model posits that the gamified app enhances user knowledge and enjoyment, which then translate into intentions and behaviors. The bold paths in Fig. 1 are our novel contributions; specifically, while we test paths from perceived value to behavioral intentions (non-bold), our main focus is on the understudied elements, the bold paths from perceived value to WOM, and from intentions to real behaviors. As indicated in Table 1, the literature has not yet fully established the impact of specific game design elements on mediator constructs such as knowledge and enjoyment, or downstream effects on outcomes (Morganti et al., 2017). To address these gaps in the literature, we hypothesize enjoyment and knowledge as mediators: game design elements → enjoyment and knowledge → WOM and energy saving behaviors.

Most efforts for social change begin by increasing the awareness of one's behaviors and how they contribute to suboptimal outcomes, to motivate behavior change. Thus, the gamified app conveys numerous facts about energy saving behaviors. In turn, knowledge is posited as contributing to more favorable behavioral intentions, thus functioning as a mediator between game design elements and subsequent consumer behaviors. Enjoyment is important as well, to ensure users continue to use the gamified app and thereby enhance their expertise about sustainability and efficient household behaviors. In addition, effects on WOM behaviors should be important in sustainability efforts as consumers learn and share in social exchanges among family members or coworkers (e.g., Paillé, Amara, & Halilem, 2018).

The gamified app, described in detail shortly, focused on three specific behaviors. One behavior was “Cool,” which was about using a fan rather than an air conditioner. “Wash” was about laundry, running full rather than partial loads, and using cold rather than hot water. “Switch” concerned turning off lights and appliances when not in use. The mediators of enjoyment and knowledge should then affect behavioral intentions and WOM:

H5. *Enjoyment should affect behavioral intentions for: (a) Cool, (b) Switch, (c) Wash.*

H6. *Enjoyment should affect WOM behaviors for: (a) Cool, (b) Switch, (c) Wash.*

H7. *Knowledge should affect behavioral intentions for: (a) Cool, (b) Switch, (c) Wash.*

H8. *Knowledge should affect WOM behaviors for: (a) Cool, (b) Switch, (c) Wash.*

Most scholars agree that behavioral intentions should be at least somewhat predictive of behavior, yet many studies do not measure ultimate outcomes, stopping with measures of behavioral intentions (Johnson et al., 2017; Morganti et al., 2017). Obviously, the link from intentions to behaviors is complex, but before giving advice to energy companies about optimizing gamified apps, it is important to examine links to actual behaviors, hence:

H9. *Behavioral intentions should affect energy saving behaviors for: (a) Cool, (b) Switch, (c) Wash.*

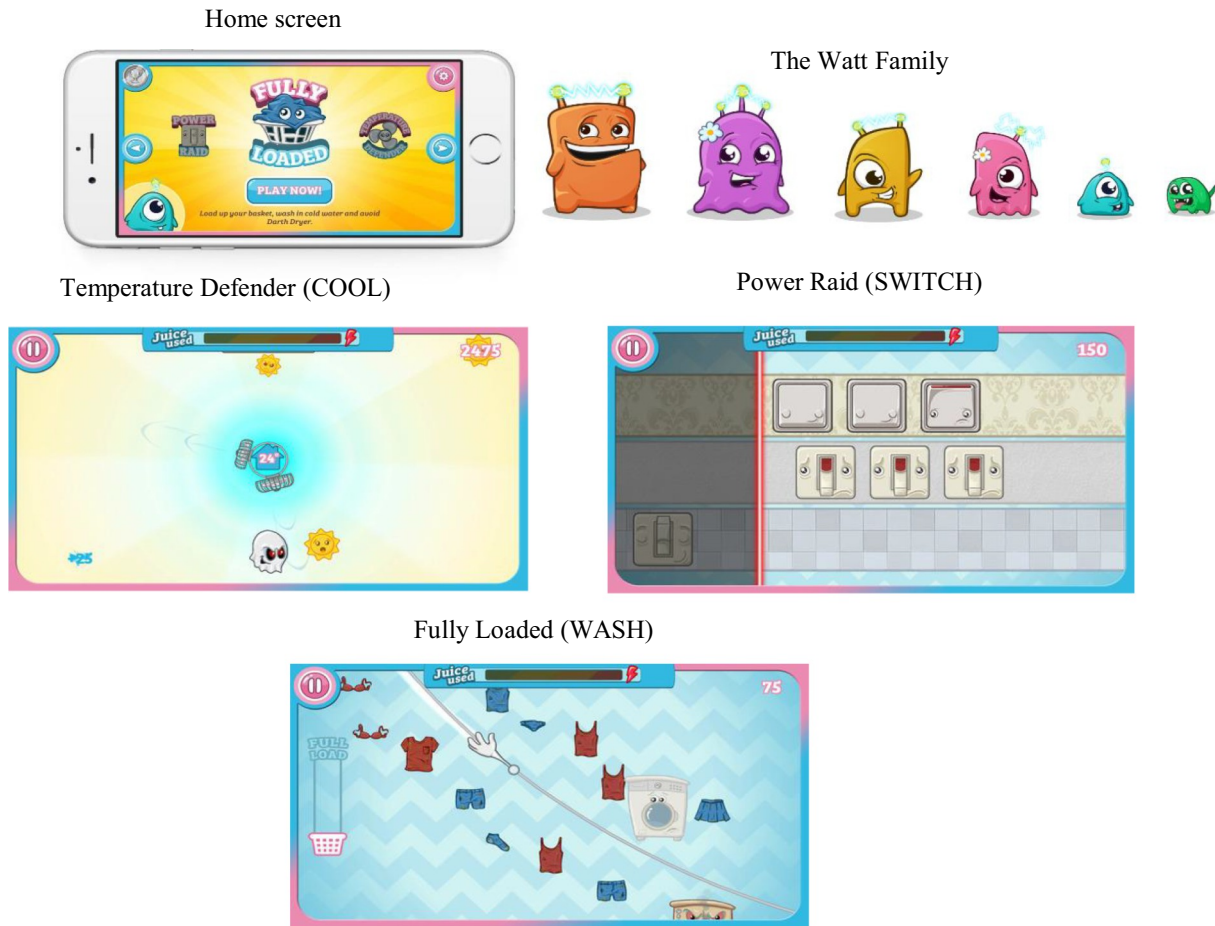


Fig. 2. Gamified app screenshots.

H10. . *Energy saving behaviors ((a) cool, (b) switch, (c) wash) should reduce household energy bills.*

Energy bill savings is measured as a comparison between pre- and post-bills (described in more detail shortly). We will also posit a moderator relationship regarding H10 examining user differences, discussed shortly.

Fig. 1 posits indirect links and hypothesized mediators, and all will be tested:

H11a. . *Game design elements will affect enjoyment and knowledge, which subsequently will affect WOM and behavioral intentions, and the latter affects energy saving behaviors.*

H11b. . *Points will affect enjoyment and knowledge through challenge.*

3.3. Hardcore versus casual users, and users versus control group

Finally, our research also examines potential differences between users based on their duration of use. Hardcore users are known for being very competitive (Scharkow et al., 2015), so points achieved or the challenge of the app may be more important. For casual users, feedback and characters are likely to be more influential because they reflect the app's engagement—specifically, feedback will help the casual user learn, and characters help casual users get involved the gamified app, thus:

H12. . *The effect of game design elements feedback and character will be stronger for casual than hardcore users, and challenge and points for hardcore.*

Finally, given the conceptualizations of sustainability as multi-faceted, we will also examine the effect of the gamified app on economic outcomes, namely the household energy bills. We will examine changes from pre- (June) to post- (September) bills and compare them to the control group.

H13. . *Gamified app users will receive greater energy bill savings than the control group who did not use the app.*

4. Method

The study comprises both subjective data (i.e., surveys) and objective data (i.e., gamified app analytics, energy bills). A sample of 1000 participants was recruited via Facebook and a third party, and screened to qualify—they had to be 18–35 years old, earning less than \$AUD50,700 (annual household income), and who rented. Of the sample, 53% were employed or students, and 42% had vocational or college degrees. Participants had to use the app for at least 10 min in total between the launch of the app in June and the closure of access to the app in September. The 10-minute requirement by the energy company quickly separated consumers who signed up for the study but then decided they were uninterested in continuing from the participants who got “hooked” on the gamified app. Another sample of customers was drawn to serve as a control group who were not exposed to the gamified app but provided pre- (June) and post- (September) survey data for comparison.

Online surveys, app analytics, and consumption data were used in this study. Online surveys were collected at two-time points, early June and late September, prior to and after the window of time during which the gamified app was accessible to users. The first survey included

Table 2

Descriptive statistics, correlations, and AVE indices (in parentheses on diagonal).

	M	SD	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
1. Points	1.96	0.62	(0.60)														
2. Challenge	4.03	0.62	0.16**	(0.54)													
3. Character	2.72	0.62	0.13*	0.22**	(0.71)												
4. Feedback	3.89	0.70	0.00	0.53**	0.21**	(0.61)											
5. Enjoyment	3.81	0.99	0.16**	0.34**	0.15**	0.15**	(0.76)										
6. Knowledge	4.29	0.60	0.15*	0.58**	0.24**	0.47**	0.26**	(0.70)									
7. Cool Intent	4.81	0.50	0.11	0.01	0.04	0.09	−0.03	0.05									
8. Wash Intent	4.75	0.68	0.06	0.09	−0.07	0.02	−0.09	0.06	0.26**								
9. Switch Intent	4.83	0.44	−0.05	0.12*	−0.03	0.07	0.02	0.14*	0.17**	0.13*							
10. Cool WOM	4.58	0.74	0.06	0.24**	0.10	0.16**	0.14*	0.21**	0.49**	0.03	0.24**						
11. Switch WOM	4.73	0.58	0.02	0.28**	0.08	0.10	0.20**	0.28**	0.08	0.01	0.54**	0.50**					
12. Wash WOM	4.72	0.55	0.07	0.25**	0.01	0.13*	0.16**	0.26**	0.14*	0.24**	0.23**	0.52**	0.54**				
13. Cool beh	2.94	0.24	−0.04	0.02	−0.04	0.03	0.01	−0.01	0.24**	0.10	0.06	0.13*	0.03	0.07			
14. Switch beh	2.94	0.26	−0.02	0.01	−0.12*	0.00	−0.05	−0.02	0.09	0.14*	0.16**	0.03	0.09	0.07	0.34**		
15. Wash beh	2.89	0.32	−0.03	0.03	−0.08	0.06	−0.11*	0.01	0.12*	0.64**	0.07	−0.04	−0.07	0.15**	0.28**	0.26**	
16. Bill Savings	\$55.44	\$177.64	0.10	0.00	−0.05	−0.10	−0.08	−0.01	0.02	0.13*	−0.05	−0.05	−0.03	−0.09	0.02	−0.02	0.70

** $p < .01$.* $p < .05$.

questions relating to energy bills used in the model (along with questions on energy saving behaviors and intentions). The post-survey included items relating specifically to the gamified app, such as game design elements, enjoyment and knowledge gained from using the app, as well as measures of energy bills, intentions, and energy saving behaviors. Of the 1000 customers recruited, 326 participants completed both surveys and had used the app for at least 10 min. Most users far exceeded this threshold; the average duration of use was 3 h, earning 701,088 points.

4.1. Gamified app

The gamified app (Fig. 2) included three mini-games, each of which related to a lower-order (i.e., specific) household energy saving behavior. The first game was called “Temperature Defender,” which focused on fan temperature settings, relating to the behavior of “Cool.” The second game was “Power Raid,” which focused on turning off lights and appliances, relating to “Switch.” The third game was “Fully Loaded,” which focused on users washing full loads of laundry, thus relating to “Wash.” All games had tips which users were encouraged to enact around the home, to transcend the app. Finally, users could pick an avatar character from the “Watt Family” to reflect household members.

4.2. Construct measurement, reliability and validity testing

Construct measures were adapted from prior literature to fit the context of the study (Appendix A). All survey items, except energy saving behaviors, were measured on a five-point scale (1 = strongly disagree to 5 = strongly agree). Energy saving behaviors were measured using a three-point scale (1 = no, 2 = not sure, 3 = yes). The dependent variable of our model, energy bill savings (\$AUD), were calculated by computing the difference between participants pre- (September) and post- (June) bills. Using app usage data, participants were classified as “casual” or “hardcore” users by a median split on gamified app use duration (which occurred at 2 h, 5 min), thus addressing a call from scholars (Hamari & Koivisto, 2015) to identify user groups using app analytics, not simply survey data, and to examine moderating factors which may impact the effect of gamification factors (Wolf, Weiger, & Hammerschmidt, 2018) such as game design elements and perceived value.

Measurement properties of the constructs were examined through tests of reliability and convergent and discriminant validity. Reliability was examined by calculating the Cronbach alphas (each met or exceeded 0.7; Appendix A). Convergent validity was assessed by computing the average variance extracted (AVE), each of which exceeded 0.6 (Table 2). Discriminant validity was assessed by showing all AVE exceeded the squared

Table 3
Hypothesis testing.

Relationship	β
Game design elements affecting user experience	
Points → Enjoyment, Knowledge, Challenge	0.17***
Points → Challenge	0.10
Points → Enjoyment	0.05
Points → Knowledge	−0.06
Feedback → Enjoyment & Knowledge	0.21***
Feedback → Enjoyment	0.08
Feedback → Knowledge	0.09*
Character → Enjoyment & Knowledge	0.34***
Character → Enjoyment	0.44***
Character → Knowledge	−0.02
Challenge → Enjoyment & Knowledge	−0.03
Challenge → Enjoyment	−0.07
Challenge → Knowledge	0.10*
User experience affecting intentions and energy saving behaviors	
Enjoyment → Behavioral Intentions	0.10*
Enjoyment → Cool Intentions	0.12*
Enjoyment → Switch Intentions	0.13*
Enjoyment → Wash Intentions	0.13*
Enjoyment → Cool WOM	0.15***
Enjoyment → Switch WOM	0.14***
Enjoyment → Wash WOM	0.17***
Knowledge → Behavioral Intentions	0.07
Knowledge → Cool Intentions	0.14*
Knowledge → Switch Intentions	0.09*
Knowledge → Wash Intentions	0.15***
Knowledge → Cool WOM	0.14***
Knowledge → Switch WOM	0.17***
Knowledge → Wash WOM	0.17***
Behavioral intentions impact on energy saving behaviors	
Behavioral Intentions → Energy save beh	0.23**
Cool Intentions → Cool energy saving behavior	0.11*
Switch Intentions → Switch energy saving behavior	0.49***
Wash Intentions → Wash energy saving behavior	0.06
Energy saving behaviors → Bill Saving	
Cool energy saving behavior → Energy Bill Saving	0.10*
Wash energy saving behavior → Energy Bill Saving	−0.04
Switch energy saving behavior → Energy Bill Saving	0.17
Cool WOM	0.21
Switch WOM	0.13
Wash WOM	0.06
Cool energy saving behaviors	0.04
Switch energy saving behaviors	0.49
Wash energy saving behaviors	0.02
Energy Bill Saving	

*** $p < .0001$.** $p < .01$.* $p < .05$.

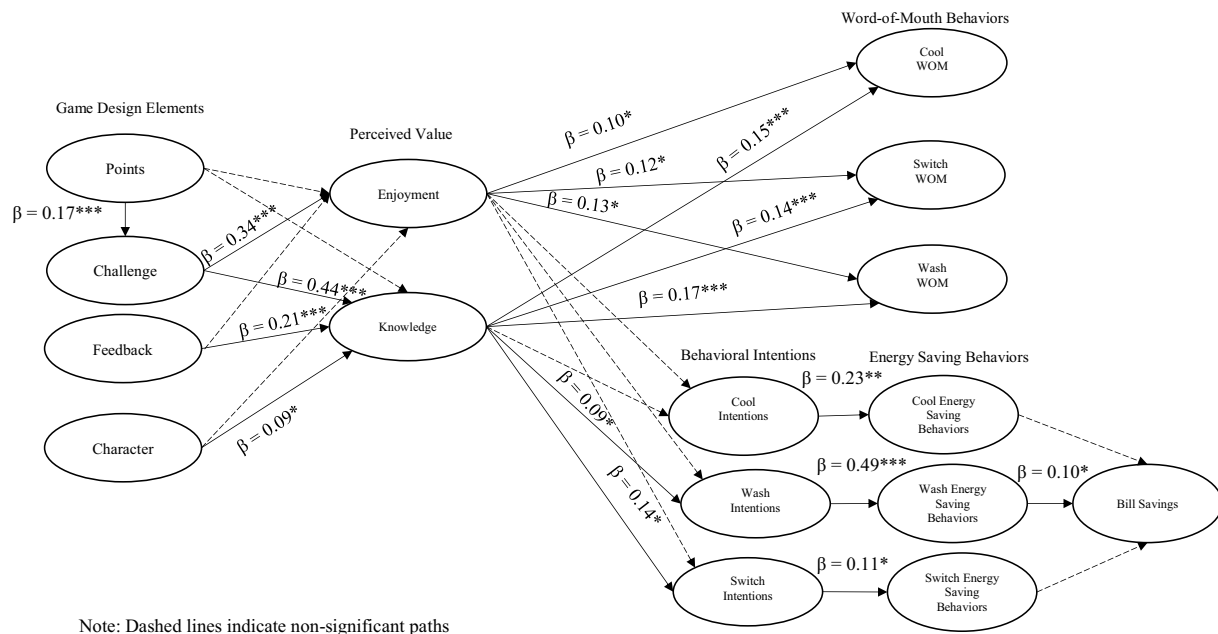


Fig. 3. Conceptual model and path coefficients.

Note: Dashed lines indicate non-significant paths.

Table 4

Mediation effects of points on knowledge and enjoyment via challenge, feedback, and character.

Points mediated by challenge	β	Points mediated by feedback	β	Points Mediated by character	β
Points $\rightarrow$ Challenge $\rightarrow$ Knowledge	0.10**	Points $\rightarrow$ Feedback $\rightarrow$ Knowledge	0.05	Points $\rightarrow$ Character $\rightarrow$ Knowledge	0.04*
Points $\rightarrow$ Challenge $\rightarrow$ Enjoyment	0.06*	Points $\rightarrow$ Feedback $\rightarrow$ Enjoyment	0.02	Points $\rightarrow$ Character $\rightarrow$ Enjoyment	0.02

** $p < .0001$.

* $p < .05$.

correlations between constructs, and results of Harman's one-factor test found the items collectively explained 28% of the variance, and a common latent factor explained only 9%, demonstrating that common method bias did not seem to contaminate the results of this study.

5. Results

The data were analyzed using PLS-SEM in SmartPLS3. A partial least squares approach to structural equations modeling is preferred when the modeling is purposed toward prediction, and is relatively robust to issues of sample size and distributions (Fornell & Bookstein, 1982). PLS-SEM is commonly used in gamified app and sustainable consumption studies (Hamari & Koivisto, 2015; Sharma & Jha, 2017).

Table 3 and Fig. 3 show that points had a significant influence on challenge, supporting H1c. Feedback significantly influences knowledge, supporting H2b, and character significantly influences knowledge, supporting H3b. Challenge significantly influences enjoyment and knowledge, supporting H4.

In the center of Table 3, results indicate that enjoyment influences WOM Cool, WOM Switch, and WOM Wash, supporting H6a–H6c. Knowledge influences intentions to Switch and intentions to Wash, supporting H7b and H7c, as well as WOM Cool, WOM Switch, and WOM Wash, supporting H8.

In the bottom of Table 3, Cool intentions influences Cool energy saving behaviors, Switch intentions influences Switch energy saving behaviors, and Wash intentions influences Wash energy saving behaviors, supporting H9. Further, Wash energy saving behaviors influences energy bill savings, supporting H10c.

5.1. Mediation effects

Table 4 shows that the game design element of “points” has significant mediation effects on enjoyment and knowledge, through the game design element of challenge. Table 5 shows that challenge had indirect effects on WOM for each of Cool, Switch, and Wash, via both the enjoyment and knowledge mediating paths (and only one effect on intentions for Switch). The game design elements of “feedback” and “character” had few indirect paths in Table 5; feedback seems to affect WOM indirectly through knowledge, but not enjoyment, and character has a single indirect effect through knowledge on Wash WOM.²

5.2. Testing user differences—the moderating effect of hardcore vs. casual users

Table 6 shows the results comparing hardcore vs. casual users. The “challenge to knowledge” relationship was stronger for hardcore than casual users, and the “feedback to enjoyment” and “character to knowledge” relationships were stronger for casual than hardcore users, as predicted in H12. The moderator of casual versus hardcore users did not differ significantly along other mediators.

5.3. Energy bill savings

We found a significant difference in pre-post energy bills for the

² We also fit a simpler model that did not distinguish between the specific energy saving behaviors (Cool, Switch, Wash) and the results (available from the authors) were consistent.

Table 5
Game design elements mediation effects on WOM and intentions via enjoyment and knowledge.

Relationship	β	Relationship	β	Relationship	β
Challenge mediated by Enjoyment		Feedback mediated by Enjoyment		Character mediated by Enjoyment	
Challenge → Enjoyment → Cool Intentions	0.00	Feedback → Enjoyment → Cool Intentions	0.00	Character → Enjoyment → Cool Intentions	0.00
Challenge → Enjoyment → Cool WOM	0.05*	Feedback → Enjoyment → Cool WOM	−0.01	Character → Enjoyment → Cool WOM	0.01
Challenge → Enjoyment → Switch Intentions	0.00	Feedback → Enjoyment → Switch Intentions	0.00	Character → Enjoyment → Switch Intentions	0.00
Challenge → Enjoyment → Switch WOM	0.06*	Feedback → Enjoyment → Switch WOM	−0.01	Character → Enjoyment → Switch WOM	0.01
Challenge → Enjoyment → Wash Intentions	−0.02	Feedback → Enjoyment → Wash Intentions	0.00	Character → Enjoyment → Wash Intentions	0.00
Challenge → Enjoyment → Wash WOM	0.06**	Feedback → Enjoyment → Wash WOM	−0.01	Character → Enjoyment → Wash WOM	0.00
Challenge mediated by Knowledge		Feedback mediated by Knowledge		Character mediated by Knowledge	
Challenge → Knowledge → Cool Intentions	0.03	Feedback → Knowledge → Cool Intentions	0.01	Character → Knowledge → Cool Intentions	0.01
Challenge → Knowledge → Cool WOM	0.08*	Feedback → Knowledge → Cool WOM	0.04*	Character → Knowledge → Cool WOM	0.02
Challenge → Knowledge → Switch Intentions	0.05*	Feedback → Knowledge → Switch Intentions	0.02*	Character → Knowledge → Switch Intentions	0.01
Challenge → Knowledge → Switch WOM	0.08***	Feedback → Knowledge → Switch WOM	0.04***	Character → Knowledge → Switch WOM	0.02
Challenge → Knowledge → Wash Intentions	0.00	Feedback → Knowledge → Wash Intentions	0.01	Character → Knowledge → Wash Intentions	0.00
Challenge → Knowledge → Wash WOM	0.09***	Feedback → Knowledge Wash WOM	0.04*	Character → Knowledge → Wash WOM	0.02*

*** $p < .0001$.

** $p < .01$.

* $p < .05$.

Table 6
Moderated effect of hardcore versus casual users.

Relationship	Casual users ($n = 187$)		Hardcore users ($n = 139$)		Difference testing	
	β	S.E.	β	S.E.	t-Statistic	Group
Points → Challenge	0.20**	0.06	0.20	0.20	0.03	ns
Points → Enjoyment	0.05	0.06	−0.27	0.28	1.52	ns
Points → Knowledge	0.10	0.06	0.10	0.09	0.02	ns
Challenge → Enjoyment	0.44***	0.08	0.32*	0.13	0.82	ns
Challenge → Knowledge	0.29**	0.09	0.61***	0.08	2.56*	Hardcore
Feedback → Enjoyment	−0.20*	0.07	0.13	0.09	2.95**	Casual
Feedback → Knowledge	0.23*	0.08	0.19**	0.06	0.20	ns
Character → Enjoyment	0.16*	0.06	−0.05	0.10	1.90	ns
Character → Knowledge	0.19**	0.07	−0.03	0.05	2.40*	Casual

ns = non-significant.

*** $p < .0001$.

** $p < .01$.

* $p < .05$.

gamified app users (June; $M = \$496.62$) to post- (September; $\$441.18$), $t(285) = 31.63$, $p < .0001$, averaging $\$55.44$ per participant or $\$221.76$ annually. Fig. 4 shows their energy bill savings (September–June) were significantly higher than the control group ($F(1, 542) = 10.89$, $p < .0001$), thus, supporting H13.

Finally, the Wash, Switch, and Cool behaviors were compared between the app participants and the control group (Fig. 5 and Table 7). With only one exception, eight of these nine tests indicated that the app enhanced consumers' energy saving behaviors, supporting H13. The results are very encouraging in terms of showing the app's effect on significantly improving energy saving behaviors and reducing household energy bills.

6. Discussion

This research was intended to offer new theoretical perspectives and applications of gamified apps in the business context of sustainable consumption of energy. A field study was used to identify benefits of gamified apps and their use for sustainable (decreased) energy consumption and monetary savings. This contribution has been achieved through examining the influence of game design elements on enjoyment and knowledge, and their influence on behavioral intentions and WOM, as well as the influence of intentions on energy saving behaviors, and on energy bill savings.

The analyses showed the ability of game design elements to

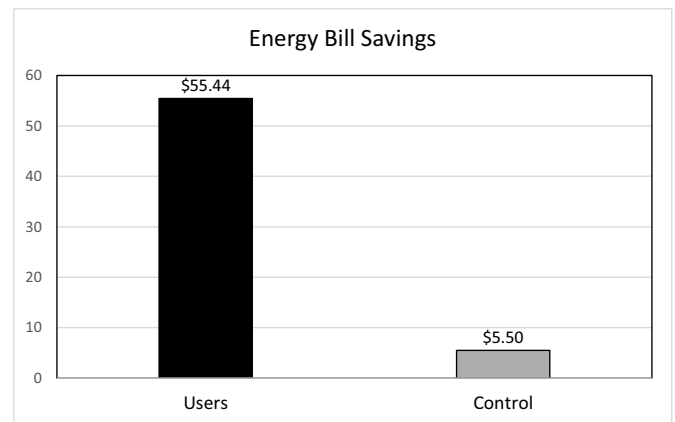


Fig. 4. Gamified app users savings significantly exceed control.

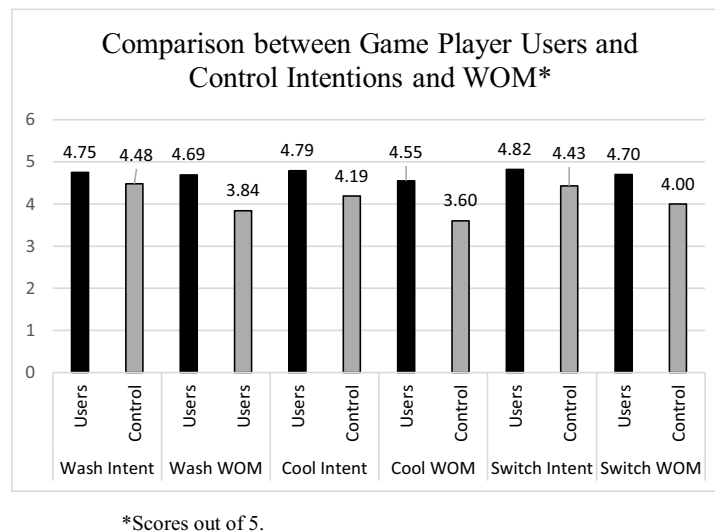
influence enjoyment and knowledge, with challenge being the most consistent influencer of each. Thus, the first aim of this study was achieved. Further, the use of specific lower-order behaviors (Cool, Wash, and Switch) revealed that enjoyment and knowledge had different effects per the second aim of this study. The results also show user groups have different responses to game design elements.

The results were quite encouraging regarding energy and environmental sustainability. In this field study, real customers demonstrated real behavioral change—energy saving changes with ambient household temperatures, laundry routines, and appliance usage, WOM to friends and family to encourage others and share more information, and of course a pronounced drop in the household energy bills as called for in the literature.

The consumer savings are a particularly important finding, an economic result of financial sustainability for a household, consistent with the environmental sustainability goals of reduced energy consumption. The results on the energy bill savings using the gamified app showed an average saving of $\$55.44$ per quarterly bill or $\$221.76$ annually, significantly higher than the control group.

6.1. Theoretical implications

First, regarding gamification, we had begun in Table 1 by surveying the literature to contrast our research with prior research. Recall Johnson et al. (2017, p.261) saying that “few studies compare the impact of differing game design elements,” thus we note our contribution in investigating effects of specific game design elements. Our research



Comparing Gamified App Users with Control Group on Energy Saving Behaviors.

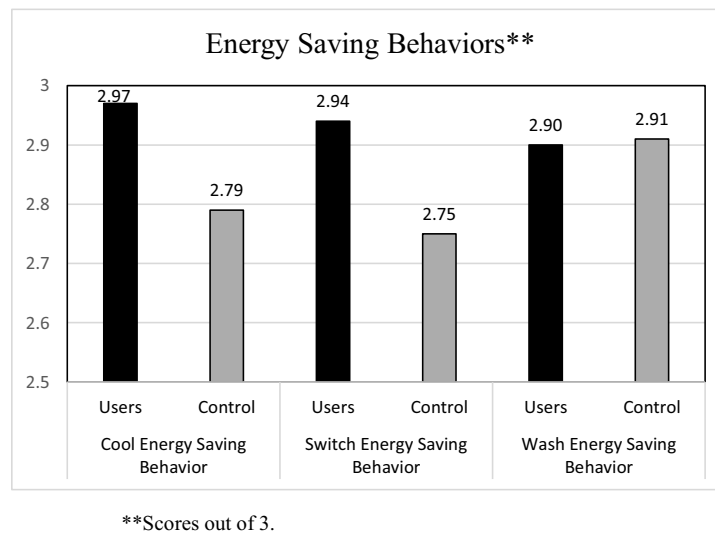


Fig. 5. Gamified app users exceed control group on behavior intentions & WOM.

Table 7
Gamified app users versus control group results.

Construct	Gamified app users	Control group	F-value
Wash intentions	4.75	4.48	10.46**
Wash WOM	4.69	3.84	99.22***
Wash energy saving behaviors ^a	2.90	2.91	0.13 ns
Switch intentions	4.82	4.43	38.91***
Switch WOM	4.70	4.00	77.20***
Switch energy saving behaviors	2.94	2.80	15.80***
Cool intentions	4.79	4.19	68.27***
Cool WOM	4.55	3.60	97.18***
Cool energy saving behaviors	2.94	2.79	19.75***

ns = non-significant.

^a Energy saving behaviors measured 1–3.

** $p < .01$.

*** $p < .0001$.

validates a model which reflects gamification definitions, specific game design elements (Deterding et al., 2011), and user value creation (Huotari & Hamari, 2017), and begins to address such calls for research into the influence of the specific game design elements on desired

outcomes (Johnson et al., 2017).

Furthermore, we theorized and showed that gamified apps can be used for sustainable consumption, namely energy usage. The app had an impact on real outcomes, not just behavioral intentions (though those were important in the model as well), which is another important contribution of our research. The existing literature has established that gamification can increase consumption and physical well-being behaviors (Hamari & Koivisto, 2015; Rodrigues et al., 2016), and we have extended the applicability to sustainable (decreased) energy consumption as well. These results further the scholarly investigation of business applications of gamified apps, with an aim to decrease (increase) consumption of behaviors which negatively (positively) impact consumer and societal well-being.

Regarding sustainability, our research advances how environmental management may move forward. Energy companies can appeal to customers in several ways—a typical educational advertising approach to “help save the planet!” may be too vague to be actionable, resulting only in discouragement and little to no pro-sustainability behaviors. Another appeal might be more specific, focusing on how consumers may benefit financially from reduced consumption. A third appeal might be fun, such as through gamification. Our research suggests that

the latter two approaches are both feasible and effective.

6.2. Practical implications

A practical implication of this research derives from the importance of game design elements. The heterogeneity across user segments showed that challenge was important for creating knowledge for hardcore users, whereas character was more influential for the casual user. These results can be used by gamification designers to more effectively impact users because the findings suggest that game design elements like character are important in early usage of a gamified app, and as usage increases, challenge becomes more important.

Managers may also benefit from offering specific lower-order behavioral recommendations. A suggestion of “turn off the lights” may be better understood than a general admonition of “help the environment,” and therefore be more empowering.

Managers and developers can also be encouraged to incorporate both enjoyment and knowledge. App designers are likely to instinctively emphasize enjoyment as part of the gamified app, yet our research shows that it can be useful for an app to be informative in showing users how to translate the app learning and implement changes in real life.

Drawing on the notion of aligned incentives, companies and governments seeking to encourage more people to engage in energy saving behaviors can consider a specific, immediate, indeed somewhat self-oriented goal, suggesting to consumers that they will see energy bill savings. To ask that customers aim to achieve “environmentally sustainable behaviors” is perhaps too long-term for customers to believe they have efficacy to properly engage and help achieve such grand goals. It may sound less altruistic to appeal to customers' financial concerns, but if doing so achieves the same large-scale environmental goal, perhaps such means should be implemented to help achieve such

important ends.

Statistics and forecasts of numerous climate changing trends are dire, with some experts even going so far as to say the earth is past the point of no return. Yet there is no utility in such gloom-saying, and it seems irresponsible to make no changes. This research shows such behaviors can be learned (or bad habits unlearned), changes can be implemented, and done so in relatively short window durations.

6.3. Limitations

While we believe that this research has some notable strengths, such as using a field study of consumers experiencing a real-world social marketing application of a gamified app and comparing aspects of their actual behaviors with a control group, limitations of this research should also be acknowledged. For example, while we tested for convergent and discriminant validity and eliminated concerns of method bias, many of our measures were self-report. Furthermore, this research investigated only one app and one consumption context, making it plausible that the results may be context-dependent. Future research could bring in more app analytics, such as GPS-locations, as well as possible consumer downsides such as concerns for privacy. Even with these open questions, we believe this research yields several theoretical and practical implications which further the fields of gamification and sustainability.

Acknowledgements

This activity has received funding from the Australian Government. The views expressed herein are not necessarily the views of the Commonwealth of Australia, and the Commonwealth does not accept responsibility for any information or advice contained herein.

Appendix A. Scales

Character^a (Hsu, Wen, & Wu, 2009) $\alpha = 0.86$

I like the characters very much and often imagine that I was the characters

I have spent much time and effort on my character

I like to play the role of the game and to do something which I cannot achieve in my real life

I try to create new personas and styles into my characters

Challenge^a (Fu et al., 2009) $\alpha = 0.86$

I enjoyed the game without feeling bored or anxious

The game provides “hints” that help me overcome the challenges

The game provides video or audio features that help me overcome the challenges

My skill gradually improved through the course of overcoming the challenges

I was encouraged by the improvement of my skills

The difficulty of the challenges increased as my skills improved

The game provides new challenges with an appropriate pacing

The game provides different levels of challenges that tailor to different players

Feedback^a (Fu et al., 2009) $\alpha = 0.84$

I received feedback on my progress in the game

I received immediate feedback on my actions

I was notified of new tasks immediately

I received information on my successes (or failures) of challenges immediately

Enjoyment^a (hedonic value items from Mathwick, Malhotra, & Rigdon, 2001) $\alpha = 0.70$

I enjoyed playing the game for its own sake, not just for assisting me in changing my behavior

I play the game for the pure enjoyment of it.

Knowledge^a (utilitarian value items adapted from Fu et al., 2009) $\alpha = 0.86$

The game increases my knowledge about saving electricity

I catch the basic ideas of the knowledge taught about saving electricity

I try to apply the knowledge in the game

The game motivates the player to integrate the knowledge taught about saving electricity

I want to know more about the facts of saving electricity

Behavioral intentions^b (Zainuddin, 2013).

How likely are you to...?

...Use a clothes line or drying rack rather than the dryer to do dry your washing
 ...Use a fan rather than the air conditioner to cool the house
 ...Switch off lights and appliances when not in use
 WOM^b (Zainuddin, 2013).

How likely are you to recommend to someone else to...?
 ...Use a fan rather than the air conditioner to cool the house
 ...Switch off lights and appliances when not in use
 ...Use a clothes line or drying rack rather than the dryer to do dry their washing
 Energy saving behaviors^b

“Watt” do you currently do in your home? Do you...
 ...Usually use a fan rather than air conditioning to cool the house
 ...Usually use cold water to do the laundry
 ...Switch off appliances to avoid standby power consumption

^a Measured for participants post only.

^b Measured pre-post for participants and control.

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